
RYAN VAN BRUINISSE

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EDUCATION

The Walker School – Marietta, GA

Date of Degree: May 2022

Cumulative GPA: 98.28

The University of California: San Diego

Degree in Cognitive Science: Machine Learning and Neural Computation (2022-6)

Minors in Mathematics and Computer Science

Current GPA: 3.554

EXPERIENCE

Relevant Coursework at UC San Diego (2022-6)

- COGS 181 - Course on the basics of neural networks, as well as recent developments in deep learning, such as deep belief nets, CNNs, RNNs, LSTMs, and reinforcement learning
- CSE 151A - Machine learning course covering knn, k-means, boosting, clustering, with a focus on the principles behind these algorithms as well as their practical uses
- MATH 152 - Focuses on the theory and applications of machine learning topics for supervised, semi-supervised, and unsupervised learning
- CSE 152A,B - Introduction to foundations, algorithms and applications of computer vision, with image formation models and deep learning for 3D construction and object detection through Python and Python libraries such as NumPy and PyPlot
- MATH 157 - Introduces the Lean coding language, which utilizes code to create proofs for mathematical statements and algorithms, investigating how it can validate AI operations and prove properties like robustness, fairness, and safety for neural networks and more
- COGS 138 - Project course focusing on data science using large datasets, with a focus on utilizing neuroscience data for exploratory data analysis
- COGS 108 - Data science fundamentals, focusing on critical skills for the industry
- DSC 40A - Introduces foundations of Data Science with fundamental topics in machine learning such as optimization, regression, empirical risk minimization, and discrete probability
- CSE 12 - Object-Oriented Design and implementation of data structures in Java and Java collections
- CSE 15L - Development of techniques in Git and Github to develop and debug software
- MATH 154 - Discrete Mathematics and Graph Theory, introducing how this logic is used in computer science
- CSE 100 - Advanced Data Structures, covering memory management, pointers, and recursion

Judicial Board Member of Warren College Student Council, UC San Diego (2024-6)

- Uphold the council's code of conduct to ensure fair and impartial resolution of student governance and disciplinary matters
- Review case materials, evaluate policy compliance, and participate in formal hearings to adjudicate disputes and alleged violations
- Collaborate with board members to interpret bylaws, draft written decisions, and recommend policy improvements to strengthen transparency and accountability
- Facilitate council meetings to discuss, debate, and approve funding requests, event organization, and new council applicants

Internship with Tally (summer 2024)

- Led the development of a computer vision algorithm to automatically track and update scores during pickleball and tennis matches using real-time video input.
- Aided in product design, making suggestions to allow implementation of such an algorithm
- Designed and implemented object detection models leveraging Hugging Face datasets, YOLO, Ultralytics, NumPy, and Torchvision to identify ball movement, court boundaries, and player actions
- Optimized system performance for low-latency processing and high accuracy across varying lighting conditions and camera angles
- Collaborated with management to test, validate, and refine the solution, resulting in a reliable automated scoring system that eliminated manual input and enhanced match analytics

Triton Baja, UC San Diego (2023-6)

- Design, develop, and maintain the team's website to showcase vehicle builds, attract sponsorships, and drive recruitment efforts
- Collaborate with engineering, marketing, and leadership teams to create a user-friendly digital presence communicating technical achievements and milestones
- Apply front-end design principles and content strategy to improve site navigation, visual branding, and sponsor visibility, directly supporting fundraising initiatives

Internship with Tealeaf AI (summers 2021-3)

- Developed predictive machine learning models leveraging internal datasets to evaluate employee efficiency and generate actionable insights
- Built and trained Convolutional Neural Networks (CNNs) and supervised learning models using PyTorch, Scikit-Learn, and NumPy
- Programmed and configured hardware systems to mine and stake Ethereum, optimizing performance and uptime
- Designed and deployed an ERC-20 cryptocurrency token on the Ethereum network to serve as a tradable internal digital asset for company use